



Institut Teknologi Bandung

Directorate of Sustainable Professional Education

CHAPTER 12

Object Detection

Boxes and labels, predicted together — a simplified YOLO built from scratch, and a pretrained detector that generalises to a pointillist painting.

Duration 2.5 hours

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Source Chollet & Watson, Deep Learning with Python 3e — chapter 12

Session Objectives

By the end of this session, participants can:

- 1 Say when to use **detection rather than segmentation**, on both compute and labelling grounds.
- 2 Distinguish **two-stage (R-CNN)** from **single-stage (YOLO, RetinaNet, SSD)** detectors, and name the trade-off.
- 3 Build a **YOLO detection head** on a pretrained backbone, and read the shape of its grid output.
- 4 Explain why the **confidence target is an IoU score** rather than a 0/1 flag.
- 5 Run a **pretrained RetinaNet**, and explain why it handles small and large objects better.
- 6 Say why single-stage detectors are **more robust to unfamiliar inputs** than two-stage ones.

BOOK

[Chapter 12 — full text](#)

NOTEBOOK

[01_coco_and_boxes.ipynb](#)

NOTEBOOK

[02_yolo_from_scratch.ipynb](#)

NOTEBOOK

[03_pretrained_retinanet.ipynb](#)

NOTEBOOK

[All chapter 12 notebooks](#)

REPO

[Author's official notebook](#)

What detection is for

Counting

How many instances of an object are in this image.

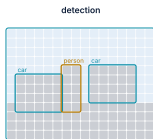
Tracking

Run detection on every frame to follow objects over time.

Cropping

Cut out the region containing an object and send a higher-resolution patch to a classifier or an OCR model.

If segmentation is a superset, why detect at all?

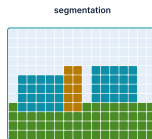


3 objects $\times$ {x, y, w, h, class}

15 numbers

17 476×

more output
for the same
question



512 $\times$ 512 pixels, one label each

262 144 labels

Both answer the same question. One of them is what the last layer has to emit — and what a person has to annotate, by hand, for every training image.

The same scene, both ways. Step through: the image, the boxes and what they cost, the mask and what it costs.

...so the rule is about what you need, not what you can get

Computational cost

A good detector typically runs **much faster** than a segmentation model.

Labelling cost

Segmentation needs **pixel-precise masks**, far more time-consuming to produce than bounding boxes.

So the rule: **always use a detector if you do not need pixel-level information** — for instance if all you want is **to count things**

01

Two-stage vs single-stage

Two broad families, and why one of them won.

Two-stage: propose, then classify

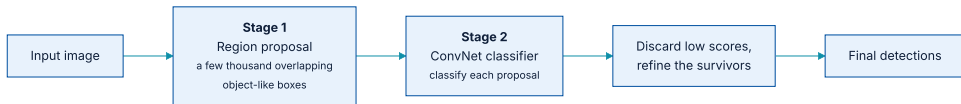


Figure 12.2 — an R-CNN extracts region proposals, then classifies each one with a ConvNet.

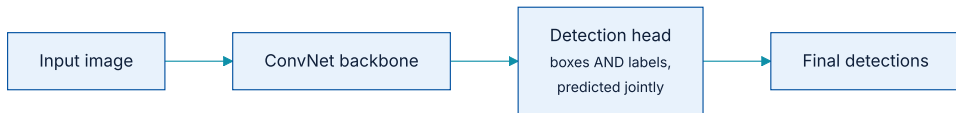
The first stage is **not very smart**: it produces boxes around areas that merely *look object-like*. The second stage decides what, if anything, is in each one.

Why that is expensive

It works well, but it requires classifying **thousands of patches for every single image**. That rules it out for most real-time applications and for embedded systems.

The book's practical take is blunt: you rarely need R-CNN at all. **With a large server GPU you are usually better off with a segmentation model** like SAM; **if you are resource-constrained you want a single-stage detector**. **Neither case points at a two-stage model.**

Single - stage: predict both at once



One model jointly predicts box coordinates and labels.

The main families are **RetinaNet**, **SSD** (Single Shot MultiBox Detector), and the **YOLO** family — *You Only Look Once*, named after the meme on purpose.

The trade-off, and the state of play

- ♦ Single-stage detectors are **significantly faster and more efficient** than two-stage ones, with a **minor potential trade-off in accuracy**.
- ♦ **YOLO is arguably the most popular object detection model there is**, especially for real-time applications.
- ♦ A new version appears roughly every year — and interestingly, **each new version tends to come from a different organisation**.

There were twelve YOLO versions at the time of writing. This chapter recreates **the original from 2015**, which is simpler to work with.

02

Training a YOLO model from scratch

COCO, a grid, and a loss with an unusual target.

An honest warning before starting

Building a detector is **an undertaking** — not because anything in it is theoretically complex, but because there is **a lot of code just to manipulate bounding boxes and predicted output**

The dataset is **COCO** — *Common Objects in Context* — one of the best-known detection datasets. It ships object labels, bounding boxes, and full segmentation masks; this chapter uses only the boxes.

Downloading COCO

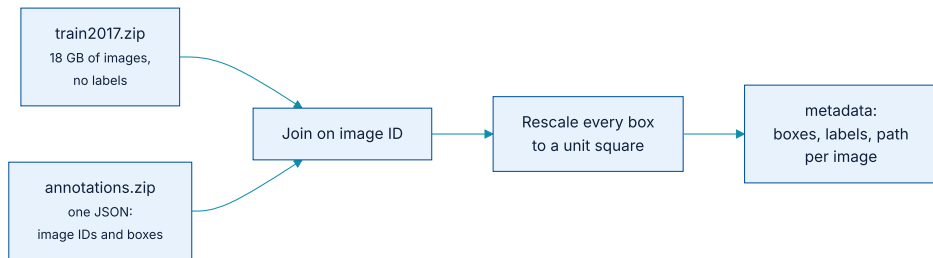
listing 12.1 – the 2017 COCO dataset

PYTHON

```
images_path = keras.utils.get_file(
    "coco",
    "http://images.cocodataset.org/zips/train2017.zip",
    extract=True,
)
annotations_path = keras.utils.get_file(
    "annotations",
    "http://images.cocodataset.org/annotations/annotations_trainval2017.zip",
    extract=True,
)
```

The first gives an **unlabelled directory of images**; the second gives all the metadata. They have to be joined.

Joining images to their boxes



COCO associates each image file with an ID, and each bounding box with one of those IDs.

...and normalising the coordinates while you are at it

listing 12.2 – parsing the annotations

PYTHON

```
with open(f"{annotations_path}/annotations/instances_train2017.json") as f:
    annotations = json.load(f)
images = {image["id"]: image for image in annotations["images"]}

def scale_box(box, width, height):
    scale = 1.0 / max(width, height)      # longest side becomes 1.0
    x, y, w, h = [v * scale for v in box]
    x += (height - width) * scale / 2 if height > width else 0    # centre the
    y += (width - height) * scale / 2 if width > height else 0    # short side
    return [x, y, w, h]
```

The two centring lines matter: images are not square, so scaling by the longest side leaves letterboxing — and the boxes have to be **shifted by half that gap** or every prediction is offset.

...and collating it per image

aggregating by image ID

PYTHON

```
metadata = {}
for annotation in annotations["annotations"]:
    id = annotation["image_id"]
    metadata.setdefault(id, {"boxes": [], "labels": []})
    image = images[id]
    metadata[id]["boxes"].append(
        scale_box(annotation["bbox"], image["width"], image["height"]))
    metadata[id]["labels"].append(annotation["category_id"])
    metadata[id]["path"] = images_path + "/train2017/" + image["file_name"]

metadata = list(metadata.values())
```

The result is one record per image carrying **its boxes, its labels, and its file path** — which is what the target-array builder consumes.

The core idea: a grid of small predictors

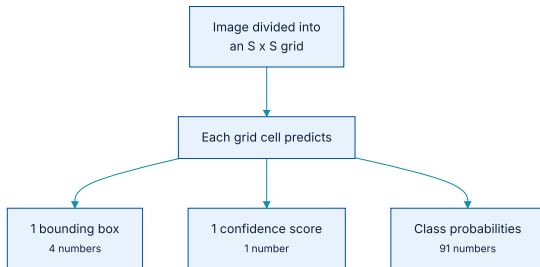


Figure 12.4 — each grid cell emits a box, a confidence, and a class distribution.

The original paper predicted several boxes per cell; this version keeps **one box per cell** for simplicity.

Why a confidence score is needed at all

So each cell also outputs a **confidence**: high where an object is detected, near zero where there is none. **Most cells should report near-zero confidence**, and that is the correct behaviour, not a failure.

The backbone: ResNet, not Xception

listing 12.5 – loading ResNet

PYTHON

```
image_size = 448

backbone = keras_hub.models.Backbone.from_preset("resnet_50_imagenet")
preprocessor = keras_hub.layers.ImageConverter.from_preset(
    "resnet_50_imagenet",
    image_size=(image_size, image_size),
)
```

- **ResNet** is structurally similar to Xception but downsamples with **strides instead of pooling** — and chapter 11 established that strides preserve location, which detection needs.
- **448 × 448** rather than 180 × 180: **input size matters a great deal** for detection, because small objects vanish at low resolution.

The detection head

listing 12.6 – attaching the head

PYTHON

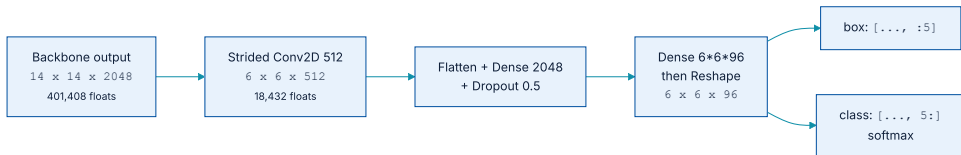
```
grid_size, num_labels = 6, 91

inputs = keras.Input(shape=(image_size, image_size, 3))
x = backbone(inputs)
x = layers.Conv2D(512, (3, 3), strides=(2, 2))(x)      # shrink the feature map
x = layers.Flatten()(x)
x = layers.Dense(2048, activation="relu", kernel_initializer="glorot_normal")(x)
x = layers.Dropout(0.5)(x)
x = layers.Dense(grid_size * grid_size * (num_labels + 5))(x)
x = layers.Reshape((grid_size, grid_size, num_labels + 5))(x)

box_predictions = x[..., :5]                          # 4 box numbers + confidence
class_predictions = layers.Activation("softmax")(x[..., 5:])
model = keras.Model(inputs, {"box": box_predictions, "class": class_predictions})
```

The `+ 5` is four box coordinates plus one confidence. Everything after that is the class distribution.

Following the shapes through the head



The strided convolution exists purely to get the float count down to something a dense layer can accept.

401,408

floats per image out of the backbone

18,432

after the strided convolution

One design choice worth noticing

Because the **entire feature map is flattened** before the dense layers, every grid detector can see features from **the whole image**. There is **no locality constraint**

That is deliberate: **large objects will not stay contained inside a single grid cell**, so a cell responsible for one needs to see beyond its own square.

77.8 M

total parameters

297 MB

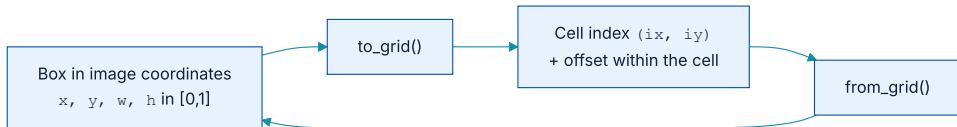
model size

Aligning the labels with the grid

The assignment rule: **each grid detector is responsible for any box whose centre falls inside its cell**. A box straddling several cells still belongs to **exactly one of them**

This is why the box format matters: the coordinates are stored as centre-x, centre-y, width, height, so the cell assignment is a direct lookup.

Two coordinate systems, and the pair of functions between them



The model predicts positions relative to a cell; the labels arrive relative to the image.

x and **y** are relative to the grid cell (0 to 1 within it), while **w** and **h** stay relative to the whole image. The widths and heights need no conversion; **only the centres do**

...written out

converting to and from the grid

PYTHON

```
def to_grid(box):
    x, y, w, h = box
    cx, cy = (x + w / 2) * grid_size, (y + h / 2) * grid_size # centre, in cells
    ix, iy = int(cx), int(cy) # which cell
    return (ix, iy), (cx - ix, cy - iy, w, h) # offset within it

def from_grid(loc, box):
    (xi, yi), (x, y, w, h) = loc, box
    x = (xi + x) / grid_size - w / 2
    y = (yi + y) / grid_size - h / 2
    return (x, y, w, h)
```

Getting one of these subtly wrong is the classic detection bug: the loss still falls, the boxes still appear, and they are **consistently in the wrong place**

Building the two target arrays

listing 12.7 – creating the targets

PYTHON

```
class_array = np.zeros((len(metadata), grid_size, grid_size))
box_array = np.zeros((len(metadata), grid_size, grid_size, 5))

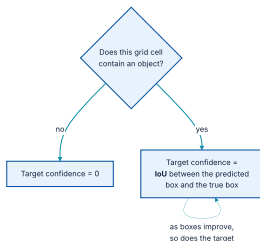
for index, sample in enumerate(metadata):
    for box, label in zip(sample["boxes"], sample["labels"]):
        (x, y, w, h) = box
        left, right = math.floor(x * grid_size), math.ceil((x + w) * grid_size)
        bottom, top = math.floor(y * grid_size), math.ceil((y + h) * grid_size)
        class_array[index, bottom:top, left:right] = label    # every cell it covers
        loc, grid_box = to_grid(box)
        box_array[index, loc[1], loc[0]] = (*grid_box, 1.0)    # only the CENTRE cell
```

Note the asymmetry: **the class map marks every cell the box overlaps**, while **the box array marks only the cell containing its centre**. Overlapping boxes are **deliberately not handled**, to keep the example readable.

Two losses, summed

OUTPUT	LOSS	NOTE
class	<code>sparse_categorical_crossentropy</code>	Entirely ordinary.
box	Sum-squared error, custom	Computed only for grid cells that actually contain a box in the labels.

The clever part: what confidence is trained against



The confidence target is not a flag — it is a measure of how good the box currently is.

The authors wanted confidence to reflect **not just the presence of an object but how good the predicted box is**. So as the model gets better at locating boxes, **the IoU rises and the learned confidence rises with it**

Computing IoU for a batch of boxes

listing 12.10 — intersection area

PYTHON

```
from keras import ops

def unpack(box):
    return box[..., 0], box[..., 1], box[..., 2], box[..., 3]

def intersection(box1, box2):
    cx1, cy1, w1, h1 = unpack(box1)
    cx2, cy2, w2, h2 = unpack(box2)
    left = ops.maximum(cx1 - w1 / 2, cx2 - w2 / 2)
    bottom = ops.maximum(cy1 - h1 / 2, cy2 - h2 / 2)
    right = ops.minimum(cx1 + w1 / 2, cx2 + w2 / 2)
    top = ops.minimum(cy1 + h1 / 2, cy2 + h2 / 2)
    return ops.maximum(0.0, right - left) * ops.maximum(0.0, top - bottom)
```

The `ops.maximum(0.0, ...)` at the end is what handles **boxes that do not overlap at all**: a negative width would otherwise produce **a spurious positive area**

Running it, and reading the prediction

listing 12.13 — visualising a prediction

PYTHON

```
x, y = next(iter(val_dataset.rebatch(1)))
preds = model.predict(x)

boxes = preds["box"][0]
classes = np.argmax(preds["class"][0], axis=-1)    # most likely label per cell

draw_prediction(path, boxes, classes, cutoff=0.1)  # a LOW cutoff, deliberately
```

The cutoff is set low because, as the next slide says plainly, **the model is not a very good detector yet**

The chapter says so out loud

Training takes **over an hour on a free Colab GPU**, and the model is **still undertrained** — the validation loss is still falling when it stops. It is starting to understand box locations and labels, **but it is not accurate**

This is worth dwelling on: the book publishes a result that does not work well, and explains why. **The purpose of the exercise is to feel how detection training behaves**, not to produce a usable COCO detector.

What would actually be needed

- 1 **Train for more epochs** — the loss had not converged.
- 2 **Use the whole COCO dataset**, not the subset.
- 3 **Data augmentation** — translating and rotating both the images **and their boxes**.
- 4 **Improve the class probability map** for overlapping boxes.
- 5 **Predict multiple boxes per grid location**, with a larger output grid.

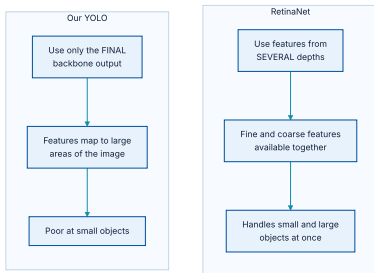
Together those approach the original YOLO training recipe. Doing it properly would take **a large amount of compute and time**, which is the honest reason to reach for a pretrained detector instead.

03

A pretrained RetinaNet

The same principles, with one important architectural difference.

The difference that matters: scale



RetinaNet uses its ConvNet differently, to handle small and large objects simultaneously.

In our YOLO we used only the **final** backbone output. Those features map to large areas of the input, so they are **not very effective at finding small objects**

Loading it, and the box format argument

loading RetinaNet

PYTHON

```
detector = keras_hub.models.ObjectDetector.from_preset(  
    "retinanet_resnet50_fpn_v2_coco",  
    bounding_box_format="rel_xywh",      # same format as our YOLO, so the same  
    )                                   # drawing utilities work unchanged  
  
predictions = detector.predict(image)
```

`rel` means **relative to the image size** — coordinates in $[0, 1]$. Most Keras models and layers that handle boxes accept this argument, and **setting it wrongly produces boxes in the wrong place with no error at all**

What comes back

inspecting the prediction

PYTHON

```
[(k, v.shape) for k, v in predictions.items()]  
# [("boxes", (1, 100, 4)),  
#  ("confidence", (1, 100)),  
#  ("labels", (1, 100)),  
#  ("num_detections", (1,))]   
  
predictions["boxes"][0][0]
```

OUTPUT

```
array([0.53, 0.00, 0.81, 0.29], dtype=float32)
```

`num_detections` is the one to read first — the arrays are padded to 100, so **the rest is not meaningful**.

Drawing the detections

listing 12.15 — running inference

PYTHON

```
fig, ax = plt.subplots(dpi=300)
draw_image(ax, path)

num_detections = predictions["num_detections"][0]
for i in range(num_detections):
    box = predictions["boxes"][0][i]
    label = predictions["labels"][0][i]
    label_name = keras_hub.utils.coco_id_to_name(label)
    draw_box(ax, box, label_name, label_to_color(label))

plt.show()
```

That is the whole of using a pretrained detector: load, predict, draw.

The pointillist painting test

RetinaNet generalises to **a pointillist painting** with ease, despite never having been trained on that style. Paintings and photographs differ enormously at the pixel level but **share a similar structure at a high level**

And the book draws an architectural conclusion from it: this is **an advantage of single-stage detectors**.

Why single-stage is more robust to novelty

Two-stage

Forced to classify **small patches in isolation** — which is extra difficult when a small patch of pixels looks nothing like the training data.

Single-stage

Can draw on features from **the entire input**, so it is more robust to novel test-time inputs.

Which connects back to chapter 5: robustness comes from

having more context to interpolate from, not from having seen this exact style before.

What has to survive this chapter

- ① **Detection when you do not need pixels** — it is cheaper to run and far cheaper to label than segmentation.
- ② **Two-stage proposes then classifies; single-stage predicts boxes and labels jointly** and is faster.
- ③ **YOLO divides the image into a grid**, and each cell predicts a box, a confidence, and a class distribution.
- ④ **Confidence is trained against IoU**, not a 0/1 flag — so it measures *how good* the box is, not only whether something is there.
- ⑤ **Input resolution matters** for detection, far more than for classification.

...and the two things worth carrying further

An honest negative result

The from-scratch model **stays undertrained**, and the chapter says so. Training a real COCO detector takes serious compute — which is the argument for pretrained models, stated with evidence.

NOTEBOOK

[03_pretrained_retinanet.ipynb](#)

NEXT

[Chapter 13 — Timeseries forecasting](#)

Robustness has a structural cause

RetinaNet handles a painting it never saw because it reads the **whole image**, and uses features from **several depths**.